

CERVIA, Italy

WMO: 161480

Lat: 44.224N

Lon: 12.307E

Elev: 6

StdP: 101.26

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.8	-2.2	-7.1	2.1	1.0	-5.1	2.5	0.9	10.7	4.8	9.4	4.5	2.9	270	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	33.1	24.1	31.9	23.9	30.5	23.5	26.5	30.1	25.6	29.4	24.7	28.8	4.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	21.1	28.7	24.3	19.3	27.8	23.2	18.0	27.0	83.0	30.5	78.6	29.8	74.9	29.1	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	7.9	6.7	-6.4	35.7	2.3	1.5	-8.1	36.8	-9.4	37.7	-10.7	38.5	-12.4	39.6		
			-6.6	28.5	2.2	1.0	-8.1	29.2	-9.4	29.8	-10.7	30.4	-12.3	31.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	4.0	5.3	9.0	12.9	17.5	21.7	24.1	24.0	19.6	15.1	9.8	4.8
	DBStd	7.70	2.72	3.00	3.14	2.73	2.71	2.80	2.32	2.51	2.84	3.03	3.54	3.13
	HDD10.0	594	188	132	55	7	0	0	0	0	0	2	46	164
	HDD18.3	2120	445	364	289	165	49	5	0	1	19	107	257	420
	CDD10.0	2062	1	1	24	93	232	351	438	433	287	161	39	2
	CDD18.3	547	0	0	0	1	22	106	180	175	56	7	1	0
	CDH23.3	5337	0	0	0	8	199	1029	1922	1768	383	28	1	0
	CDH26.7	1857	0	0	0	1	31	316	743	686	78	1	0	0
Wind	WSAvg	2.8	2.7	2.8	3.0	2.9	2.7	2.7	2.9	2.8	2.7	2.5	2.7	2.8
Precipitation	PrecAvg	680	39	44	56	61	52	48	35	49	77	80	77	56
	PrecMax	957	109	135	127	115	146	131	108	100	159	171	158	173
	PrecMin	471	3	3	9	22	13	8	1	2	26	10	13	7
	PrecStd	142	29	31	34	26	32	29	24	31	38	45	41	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	16.6	20.8	24.1	29.8	33.1	34.2	35.0	31.0	26.1	21.1	15.8
		MCWB	11.0	11.7	13.7	17.5	20.6	23.4	24.0	24.3	23.3	20.4	16.2	12.5
	2%	DB	12.0	14.1	18.8	22.0	27.1	31.4	33.0	33.2	28.9	23.9	18.2	12.9
		MCWB	8.9	9.9	13.1	16.3	19.6	23.6	24.1	24.2	22.9	19.3	15.3	10.6
	5%	DB	9.9	12.2	16.9	20.2	25.7	30.0	31.8	32.0	27.0	22.0	16.9	10.9
		MCWB	8.1	9.2	12.2	15.3	19.5	23.1	23.8	24.1	21.6	18.7	14.9	9.1
	10%	DB	8.2	10.8	15.1	18.9	23.9	28.3	30.2	30.2	25.2	20.2	15.1	9.2
		MCWB	6.8	8.3	11.4	14.7	18.3	22.0	23.4	23.9	20.5	17.5	13.6	8.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	12.4	15.3	19.2	22.9	27.0	28.0	28.0	25.9	21.9	17.6	13.1
		MCDB	13.6	15.4	18.5	22.3	26.1	30.2	30.9	30.5	28.5	24.0	19.1	14.9
	2%	WB	9.7	10.8	13.9	17.6	21.4	25.4	26.5	26.6	24.1	20.4	16.1	11.0
		MCDB	11.2	13.0	17.0	20.5	25.0	29.2	30.0	30.2	27.2	22.4	17.8	12.6
	5%	WB	8.2	9.8	12.9	16.3	20.3	24.2	25.5	25.6	22.6	19.2	15.1	9.6
		MCDB	9.5	11.7	15.9	19.0	23.9	28.1	29.4	29.7	25.7	21.2	16.5	10.6
	10%	WB	7.0	8.7	11.8	15.2	19.3	23.1	24.5	24.7	21.4	18.0	13.8	8.2
		MCDB	7.8	10.4	14.5	17.9	22.8	27.0	28.7	28.9	24.3	20.0	15.2	9.1
Mean Daily Temperature Range	5% DB	MDBR	6.8	8.7	9.8	9.8	10.4	10.8	11.1	11.0	10.1	8.4	6.8	6.5
		MCDBR	9.9	11.6	12.7	12.4	12.8	12.5	12.6	12.8	11.7	10.5	8.4	8.8
		MCWBR	7.6	8.3	8.2	7.9	7.3	6.8	6.9	7.1	6.8	7.0	6.4	6.8
	5% WB	MCDBR	8.6	10.5	11.3	11.0	11.3	11.5	11.3	11.7	10.4	9.6	7.4	7.7
		MCWBR	7.1	8.0	8.0	7.8	7.5	7.2	7.1	7.4	7.0	7.1	6.1	6.5
Clear-Sky Solar Irradiance	taub	0.355	0.375	0.408	0.442	0.441	0.449	0.441	0.434	0.422	0.411	0.379	0.351	
	taud	2.403	2.321	2.237	2.169	2.221	2.244	2.272	2.308	2.317	2.324	2.346	2.431	
	Ebn at Noon	753	801	820	822	834	827	829	821	799	750	710	719	
	Edh at Noon	78	99	122	141	138	136	130	122	112	98	81	69	
All-Sky Solar Radiation	RadAvg	1.34	2.20	3.49	4.64	5.79	6.58	6.72	5.75	4.19	2.63	1.51	1.19	
	RadStd	0.20	0.33	0.44	0.42	0.46	0.34	0.31	0.36	0.29	0.28	0.18	0.16	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	+1.78	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (19 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	-90	+85	+46	

Nomenclature: See separate page